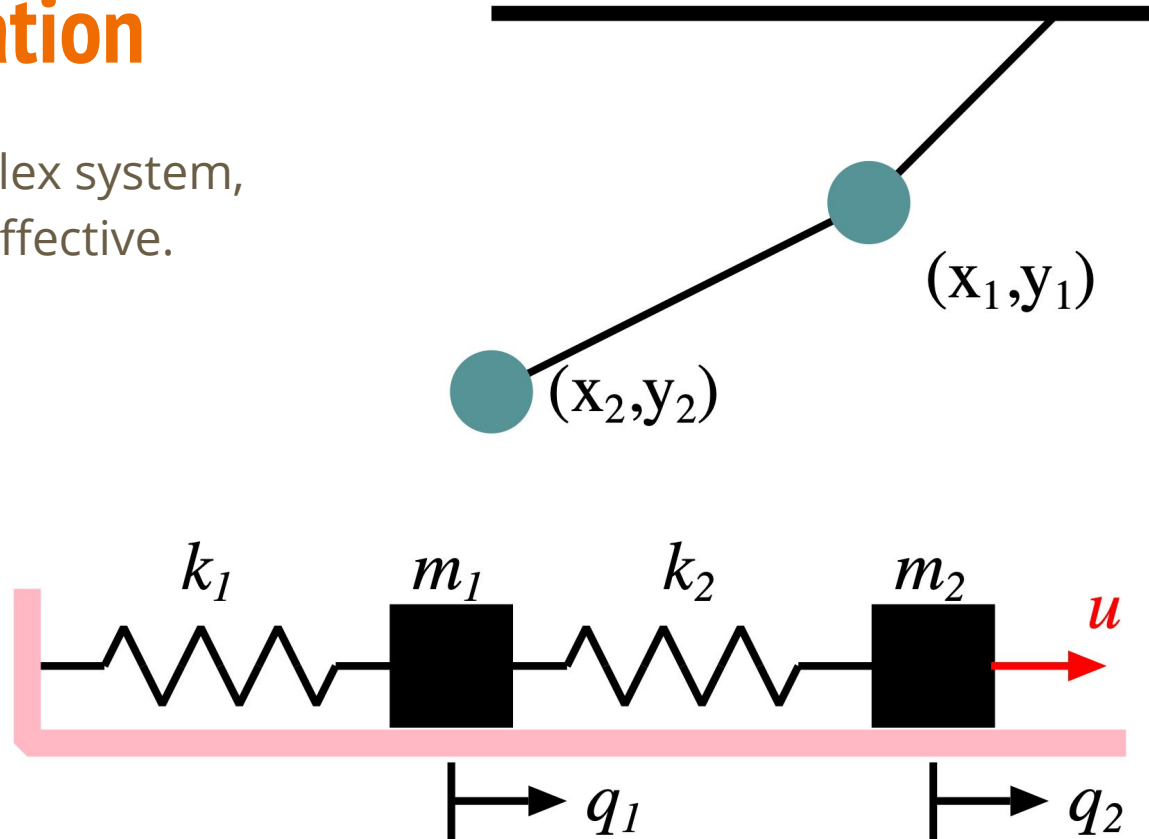

Robot Arm 2

— Electric, Electronic and System
Engineering —

Lagrange Method

Lagrange Equation

For modeling of complex system, lagrange equation is effective.



Lagrange Equation of Vertical Robot Arm

$$L = K - P$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \tau_i$$

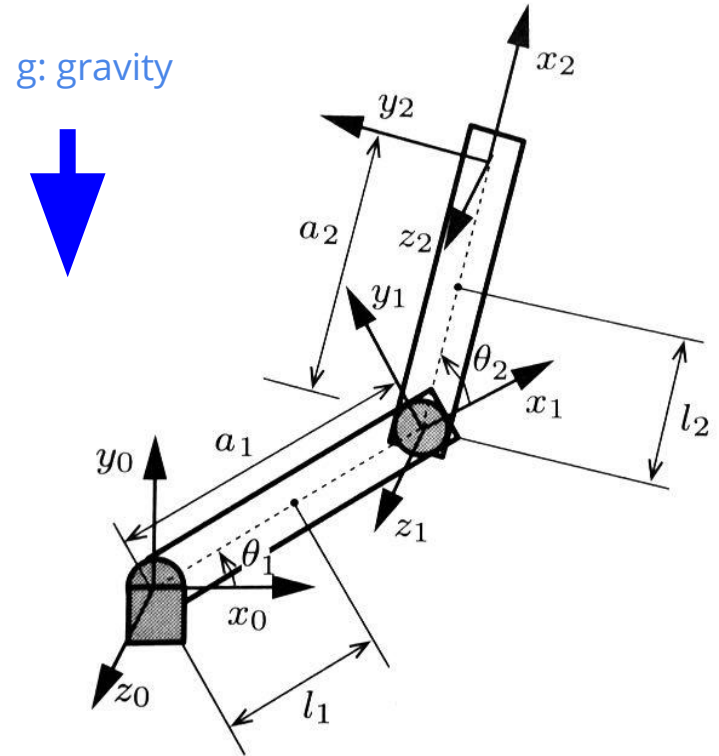
L: Lagrangean

K: Kinetic Energy

P: Potential Energy

τ_i : Generalized Input

q_i : Generalized Position



Energy of Link 1

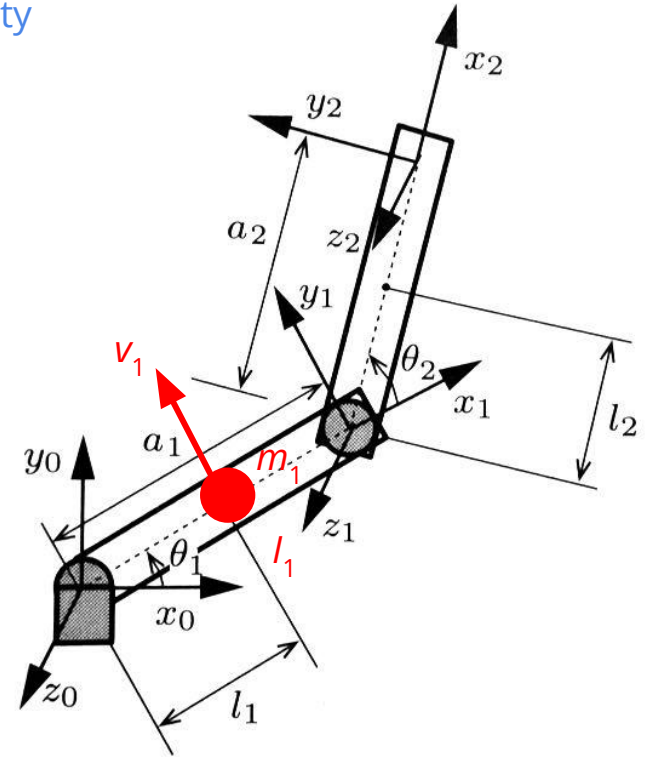
Kinetic Energy:

$$K_1 = \frac{1}{2} \text{?} + \frac{1}{2} \text{?}$$

Potential Energy:

$$P_1 = \text{?}$$

g: gravity



Energy of Link 2

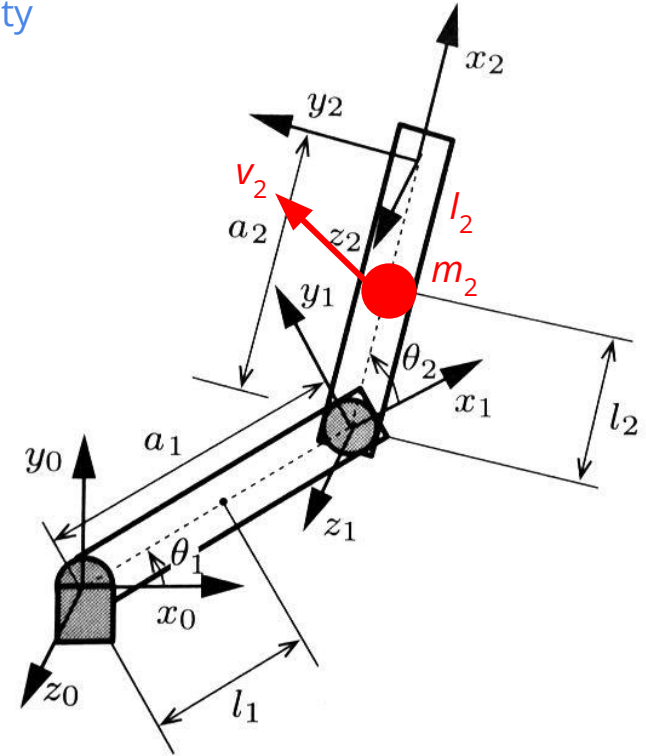
Kinetic Energy:

$$K_2 = \frac{1}{2}m_2v_2^2 + \frac{1}{2}I_2(\dot{\theta}_1 + \dot{\theta}_2)^2$$

Potential Energy:

$$P_2 = m_2g(a_1\sin\theta_1 + l_2\sin(\theta_1 + \theta_2))$$

g: gravity



Lagrangian

$$\begin{aligned} L &= K_1 + K_2 - P_1 - P_2 \\ &= \frac{1}{2} I_1 \dot{\theta}_1^2 + \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} I_2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \\ &\quad + \frac{1}{2} m_2 \left(a_1^2 \dot{\theta}_1^2 + 2a_1 l_2 \cos \theta_2 \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) + l_2^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \right) \\ &\quad - m_1 g l_1 \sin \theta_1 - m_2 g (a_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2)) \end{aligned}$$

Lagrange Equation: Link 1

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} = \tau_1$$

$$\frac{\partial L}{\partial \dot{\theta}_1} = I_1 \dot{\theta}_1 + m_1 l_1^2 \dot{\theta}_1 + I_2 \dot{\theta}_1 + I_2 \dot{\theta}_2 + m_2 a_1^2 \dot{\theta}_1 + \dots$$

$$\frac{\partial L}{\partial \theta_1} = -m_1 g l_1 \cos \theta_1 - m_2 g a_1 \cos \theta_1 - \dots$$

Lagrange Equation: Final Form

$$M(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + C(\dot{\boldsymbol{\theta}}, \boldsymbol{\theta}) + G(\boldsymbol{\theta}) = \boldsymbol{\tau}$$

$$M(\boldsymbol{\theta}) = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$$

$$M_{11} = I_1 + I_2 + m_1 l_1^2 + m_2(a_1^2 + 2a_1 l_2 \cos\theta_2 + l_2^2)$$

$$M_{12} = M_{21} = I_2 + m_2(a_1 l_2 \cos\theta_2 + l_2^2)$$

$$M_{22} = I_2 + m_2 l_2^2$$

$$C(\dot{\boldsymbol{\theta}}, \boldsymbol{\theta}) = \begin{bmatrix} -m_2 a_1 l_2 \sin\theta_2 (2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) \\ m_2 a_1 l_2 \sin\theta_2 \dot{\theta}_1^2 \end{bmatrix}$$

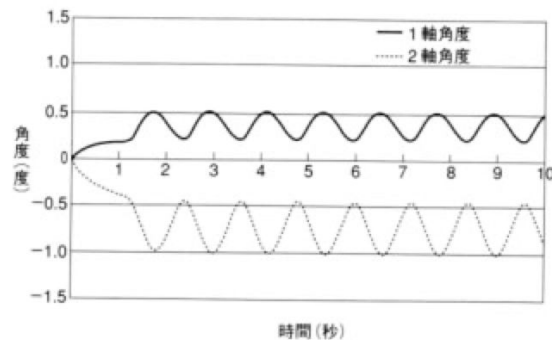
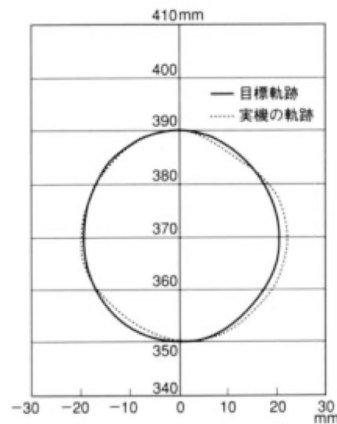
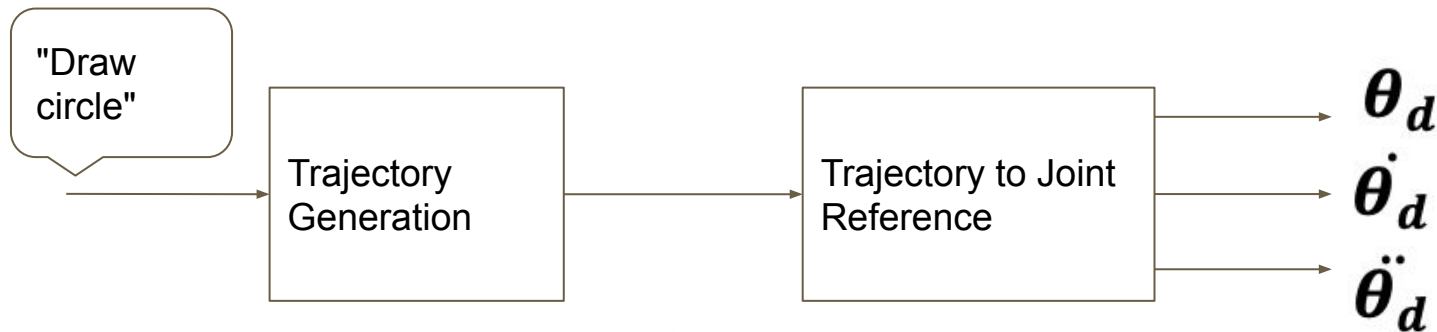
coriolis force (moment)

$$G(\boldsymbol{\theta}) = \begin{bmatrix} m_1 g l_1 \cos\theta_1 + m_2 g (a_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2)) \\ m_2 g l_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

gravitational force (moment)

Trajectory Following Control of Robot Arm

Task Command to Joint Reference



Desired Torque Input to Robot Arm

We decide the torque input as follows;

$$\tau_d = C(\dot{\theta}, \theta) + G(\theta) + M(\theta)(\ddot{\theta}_d + D(\dot{\theta}_d - \dot{\theta}) + K(\theta_d - \theta))$$

to Robot Arm described as;

$$M(\theta)\ddot{\theta} + C(\dot{\theta}, \theta) + G(\theta) = \tau$$

Then, we got;

$$M(\theta)\ddot{\theta} + \boxed{C(\dot{\theta}, \theta) + G(\theta)} = \boxed{C(\dot{\theta}, \theta) + G(\theta)} + M(\theta)(\ddot{\theta}_d + D(\dot{\theta}_d - \dot{\theta}) + K(\theta_d - \theta))$$

actual calculated

Error of Robot Arm Joint Angle

From the above equation, when calculation is done suitably,

$$M(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} = M(\boldsymbol{\theta})(\ddot{\boldsymbol{\theta}}_d + D(\dot{\boldsymbol{\theta}}_d - \dot{\boldsymbol{\theta}}) + K(\boldsymbol{\theta}_d - \boldsymbol{\theta}))$$

$$\mathbf{e} \equiv \boldsymbol{\theta}_d - \boldsymbol{\theta}$$

$$\ddot{\mathbf{e}} + D\dot{\mathbf{e}} + K\mathbf{e} = 0$$

Error goes to 0, if D, K are defined suitably.

Block Diagram of Robot Arm Control

measured:

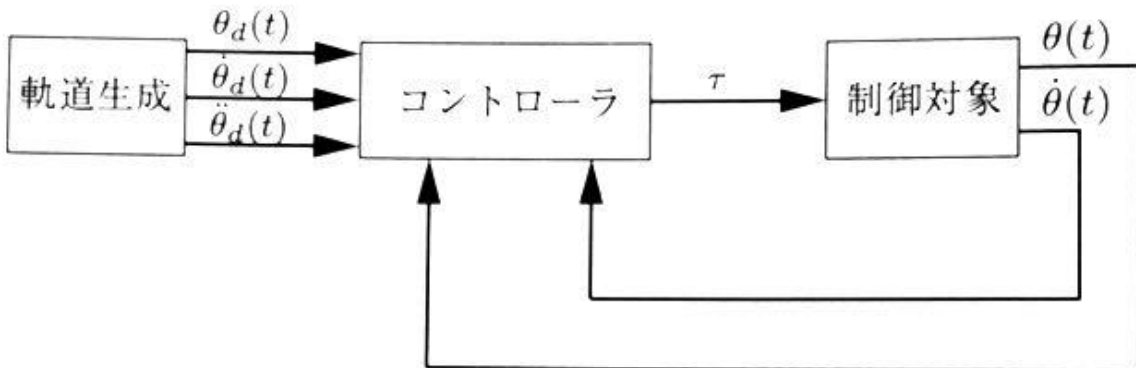
Joint Angle

Joint Angular Speed

not measured:

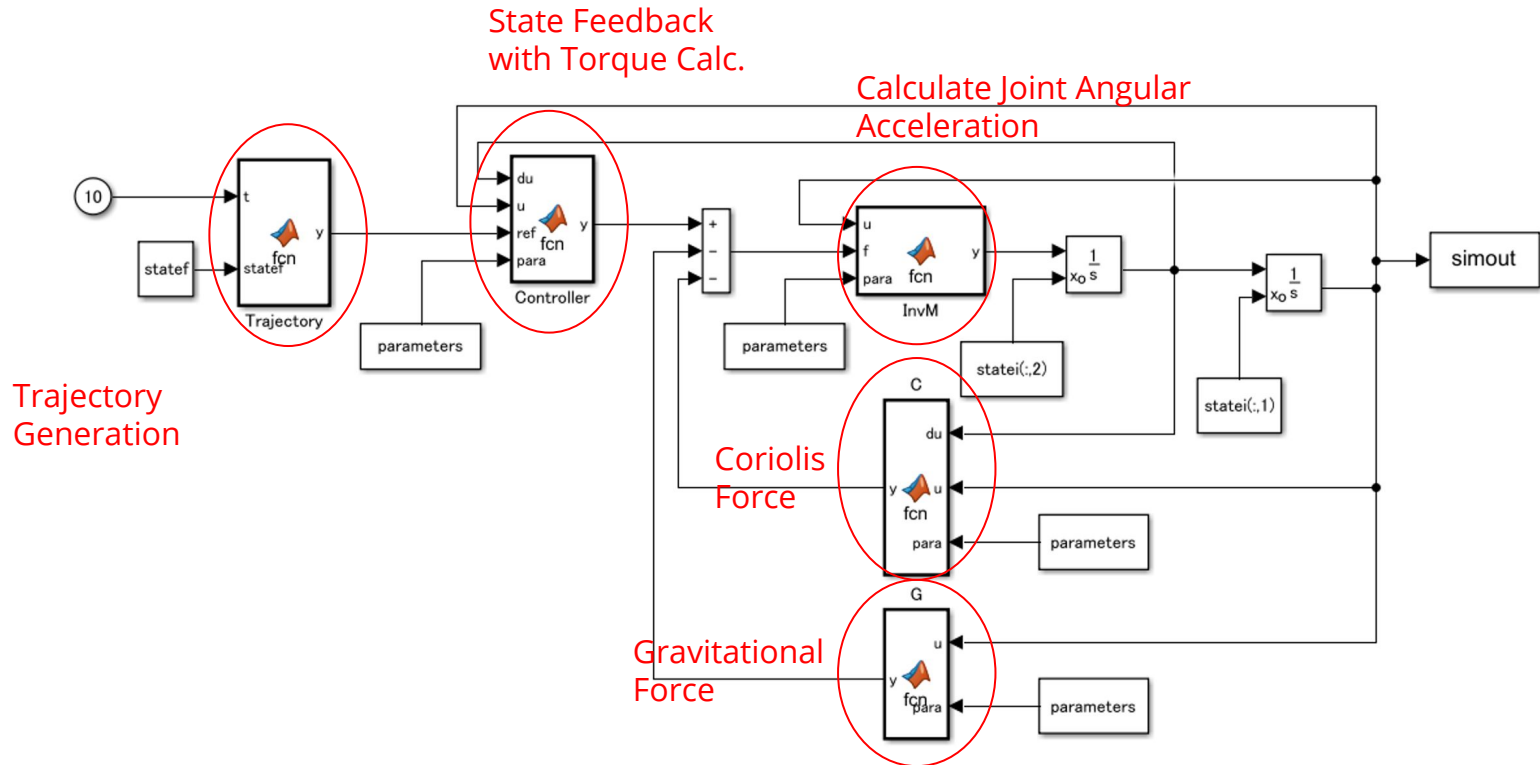
Joint Angular Acceleration

→difficult to measure precisely



MATLAB/Simulink Program

Total System

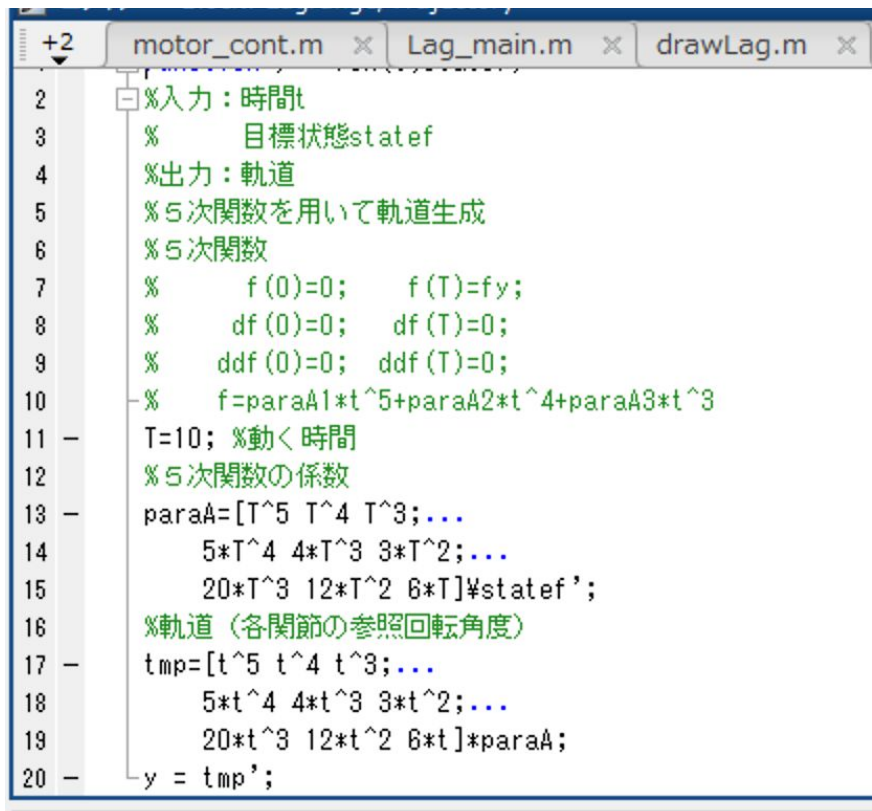


About Trajectory.m

Make a trajectory using 5th order function.

Start: velocity & acceleration = 0
position = 0

Goal: velocity & acceleration = 0
position = fy

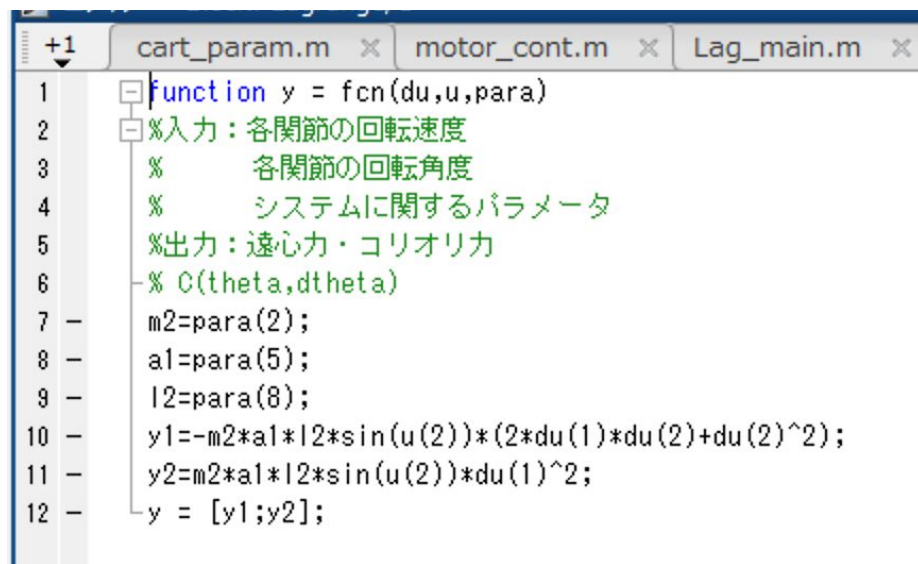


```
+2 motor_cont.m x Lag_main.m x drawLag.m x
2 %入力：時間
3 % 目標状態statef
4 %出力：軌道
5 %5次関数を用いて軌道生成
6 %5次関数
7 % f(0)=0; f(T)=fy;
8 % df(0)=0; df(T)=0;
9 % ddf(0)=0; ddf(T)=0;
10 % f=paraA1*t^5+paraA2*t^4+paraA3*t^3
11 - T=10; %動く時間
12 %5次関数の係数
13 - paraA=[T^5 T^4 T^3;...
14 5*T^4 4*T^3 3*T^2;...
15 20*T^3 12*T^2 6*T]*statef';
16 %軌道（各関節の参照回転角度）
17 - tmp=[t^5 t^4 t^3;...
18 5*t^4 4*t^3 3*t^2;...
19 20*t^3 12*t^2 6*t]*paraA;
20 - y = tmp';
```

About C.m

Calculate coriolis force using joint angle & angular speed.

$$C(\dot{\theta}, \theta) = \begin{bmatrix} -m_2 a_1 l_2 \sin \theta_2 (2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) \\ m_2 a_1 l_2 \sin \theta_2 \dot{\theta}_1^2 \end{bmatrix}$$

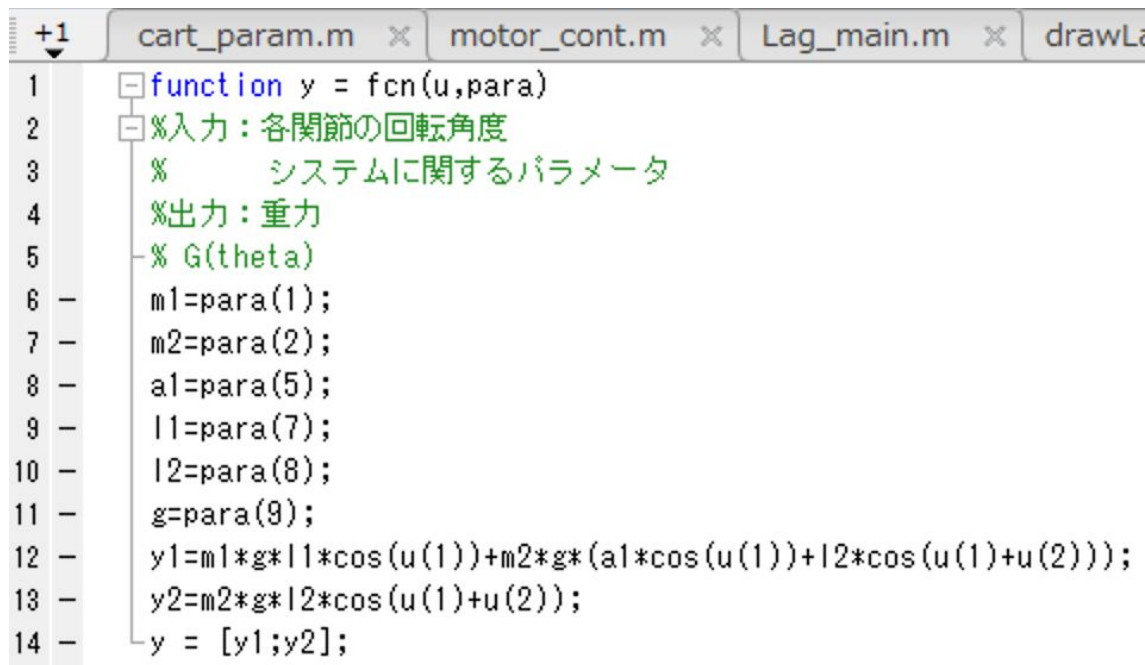


The screenshot shows a MATLAB script named `cart_param.m` with the following content:

```
1 function y = fcn(du,u,para)
2 %入力：各関節の回転速度
3 %   各関節の回転角度
4 %   システムに関するパラメータ
5 %出力：遠心力・コリオリ力
6 % C(theta,dtheta)
7 m2=para(2);
8 a1=para(5);
9 l2=para(8);
10 y1=-m2*a1*l2*sin(u(2))*(2*du(1)*du(2)+du(2)^2);
11 y2=m2*a1*l2*sin(u(2))*du(1)^2;
12 y = [y1;y2];
```

About G.m

Calculate
gravitational force
using joint angle.



```
+1 cart_param.m x motor_cont.m x Lag_main.m x drawLa
1 function y = fcn(u,para)
2 %入力：各関節の回転角度
3 % システムに関するパラメータ
4 %出力：重力
5 % G(theta)
6 m1=para(1);
7 m2=para(2);
8 a1=para(5);
9 l1=para(7);
10 l2=para(8);
11 g=para(9);
12 y1=m1*g*l1*cos(u(1))+m2*g*(a1*cos(u(1))+l2*cos(u(1)+u(2)));
13 y2=m2*g*l2*cos(u(1)+u(2));
14 y = [y1;y2];
```

$$G(\theta) = \begin{bmatrix} m_1 g l_1 \cos \theta_1 + m_2 g (a_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2)) \\ m_2 g l_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

About InvM.m

Calculate joint angular acceleration using matrix M and torque input.

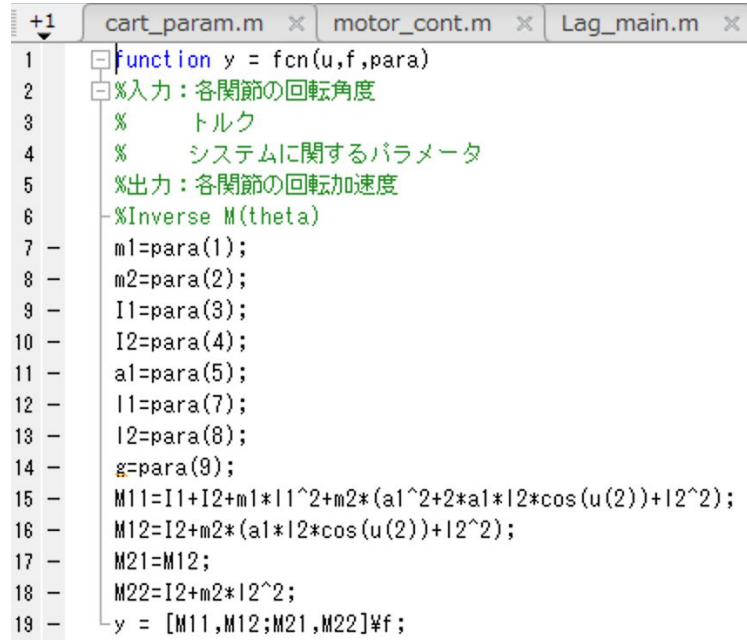
$$\ddot{\theta} = M(\theta)^{-1}(\tau - C(\dot{\theta}, \theta) - G(\theta))$$

$$M(\theta) = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$$

$$M_{11} = I_1 + I_2 + m_1 l_1^2 + m_2(a_1^2 + 2a_1 l_2 \cos \theta_2 + l_2^2)$$

$$M_{12} = M_{21} = I_2 + m_2(a_1 l_2 \cos \theta_2 + l_2^2)$$

$$M_{22} = I_2 + m_2 l_2^2$$



```
+1 cart_param.m x motor_cont.m x Lag_main.m x
1 function y = fcn(u,f,para)
2 %入力：各関節の回転角度
3 % トルク
4 % システムに関するパラメータ
5 %出力：各関節の回転加速度
6 %Inverse M(theta)
7 m1=para(1);
8 m2=para(2);
9 I1=para(3);
10 I2=para(4);
11 a1=para(5);
12 l1=para(7);
13 l2=para(8);
14 g=para(9);
15 M11=I1+I2+m1*l1^2+m2*(a1^2+2*a1*l2*cos(u(2))+l2^2);
16 M12=I2+m2*(a1*l2*cos(u(2))+l2^2);
17 M21=M12;
18 M22=I2+m2*l2^2;
19 y = [M11,M12;M21,M22]*f;
```

About Controller.m

Calculate desired torque input.

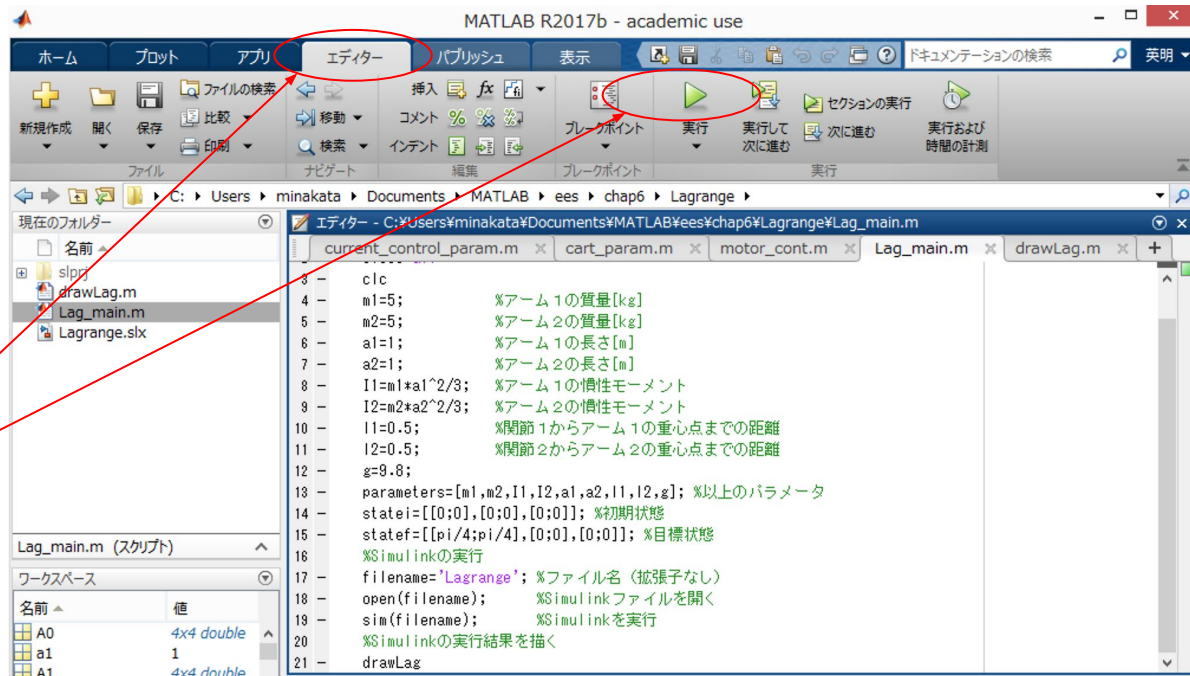
```
+1 cart_param.m x motor_cont.m x Lag_main.m x drawLag
14 - g=para(9);
15 - D=100*eye(2); %微分制御のゲイン
16 - K=100*eye(2); %比例制御のゲイン
17 %慣性力の係数
18 - M11=I1+I2+m1*l1^2+m2*(a1^2+2*a1*l2*cos(u(2))+l2^2);
19 - M12=I2+m2*(a1*l2*cos(u(2))+l2^2);
20 - M21=M12;
21 - M22=I2+m2*l2^2;
22 - qref2dot=ref(:,3); %参照回転加速度
23 - qrefdot=ref(:,2); %参照回転速度
24 - qref=ref(:,1); %参照回転角度
25 %遠心力・コリオリ力
26 - C1=-m2*a1*l2*sin(u(2))*(2*du(1)*du(2)+du(2)^2);
27 - C2=m2*a1*l2*sin(u(2))*du(1)^2;
28 %重力
29 - G1=m1*g*l1*cos(u(1))+m2*g*(a1*cos(u(1))+l2*cos(u(1)+u(2)));
30 - G2=m2*g*l2*cos(u(1)+u(2));
31 - y = [C1;C2]+[G1;G2]+...
32 [M11,M12;M21,M22]*(qref2dot+D*(qrefdot-du)+K*(qref-u));
```

$$\tau_d = C(\dot{\theta}, \theta) + G(\theta) + M(\theta)(\ddot{\theta}_d + D(\dot{\theta}_d - \dot{\theta}) + K(\theta_d - \theta))$$

Lag_main.m Controls this simulation

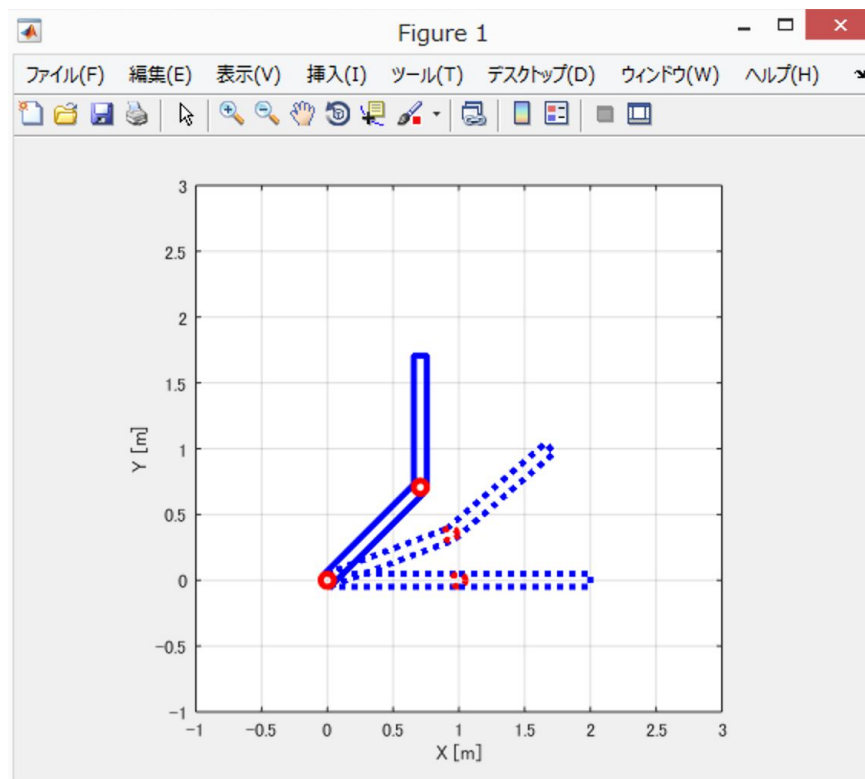
Lag_main.m controls robot arm simulation.

1. open 'Lag_main.m'
2. move to 'editor' tab
3. click 'run'



Example Output

Robot Arm rise up.



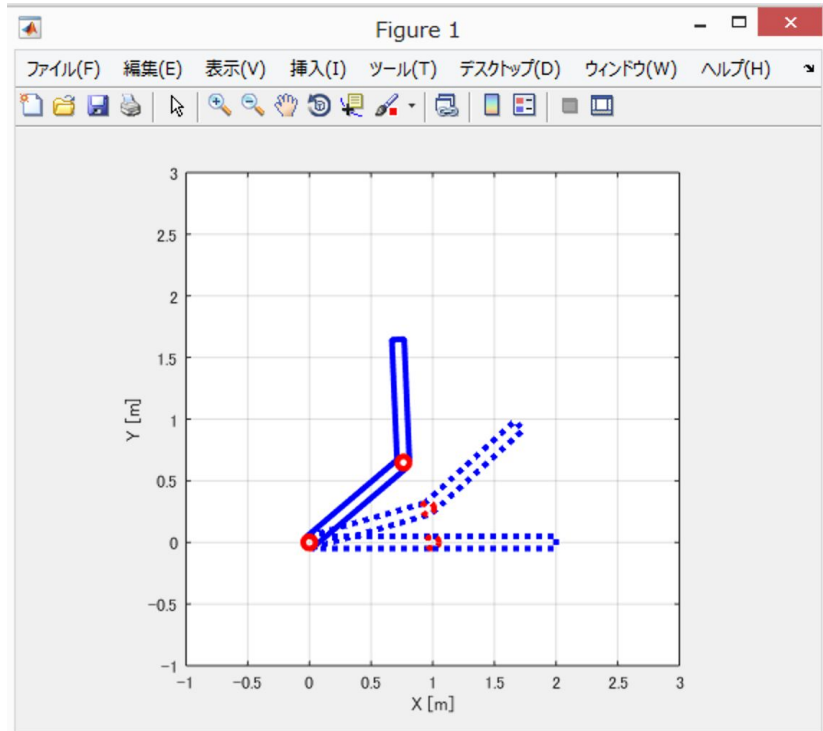
When Torque Calculation is NOT Suitable...

If we use torque input as following;

$$\tau_d = M(\theta)(\ddot{\theta}_d + D(\dot{\theta}_d - \dot{\theta}) + K(\theta_d - \theta))$$

controller gain D&K become very important to the performance of Robot Arm.

D=100, K=100



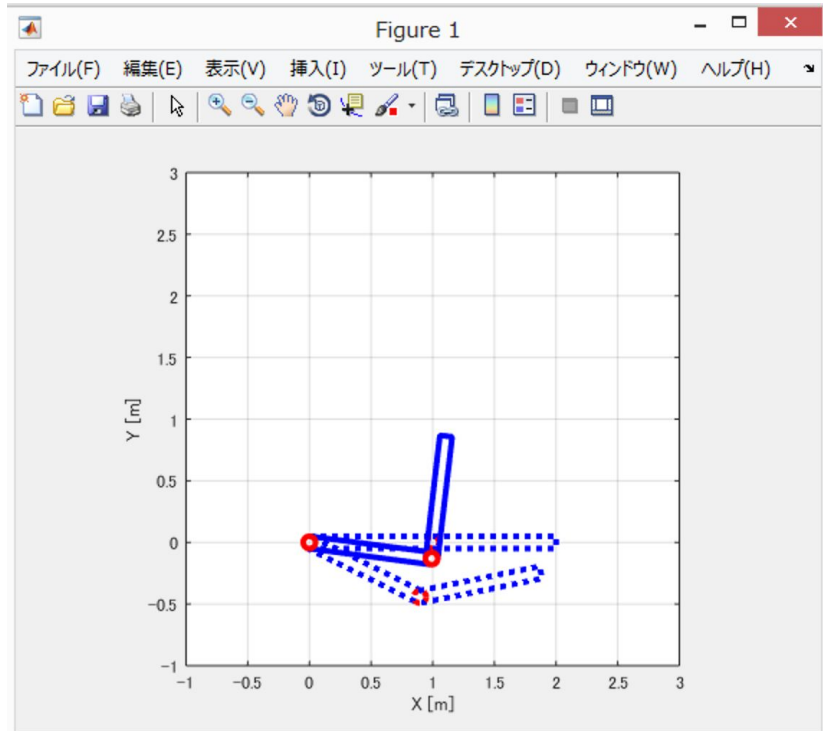
When Torque Calculation is NOT Suitable...

If we use torque input as following;

$$\tau_d = M(\theta)(\ddot{\theta}_d + D(\dot{\theta}_d - \dot{\theta}) + K(\theta_d - \theta))$$

controller gain D&K become very important to the performance of Robot Arm.

D=10, K=10



When Torque Calculation is Suitable...

If we use original torque input,
controller gain D&K can be reduced.

$D=10$, $K=10$

The performance of Robot Arm seems
good.

